REDUCEDNESS OF AN ALGEBRAIC SPACE DETECTED ON LOCAL RINGS, WITH GLOBAL CONSEQUENCES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file spaces-properties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 13 S10-geom published cards for spaces-properties.tex (prop-occupied.json, 2026-10-14); those 13 labels are excluded from this batch by design. For the 77 free labels: 0 occupied. The registry mentions field (7557 files) and focused-seeds (645) both point to the same S10-geom cards (dual-source check), no other occupation.

Source reference. Original remark 0BBL: sources/stacks-chow/tags.txt, source lines 14249–14249. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01OL: sources/stacks-chow/tags.txt, source lines 2198–2198. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let S be a scheme. Let X be an algebraic space over S. The following are equivalent*

(1) *X is reduced,*
(2) *for every $x \in |X|$ the local ring of X at x is reduced (Remark 0BBL).*

In this case $\Gamma(X, \mathcal{O}_X)$ is a reduced ring and if $f \in \Gamma(X, \mathcal{O}_X)$ has $X = V(f)$, then $f = 0$.

Proof. The equivalence of (1) and (2) follows from Properties, Lemma 01OL applied to affine schemes étale over X. The final statements follow the cited lemma and fact that $\Gamma(X, \mathcal{O}_X)$ is a subring of $\Gamma(U, \mathcal{O}_U)$ for some reduced scheme U étale over X. □

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